

Natural Language Processing Exam Evaluation

25/26

The final grade will depend on two outcomes:

1. **Written test** — open and closed questions or exercises related to concepts explained in class. **Mandatory to take - maximum grade 28/30**
2. **Homework (HW)** or extending HW as a project (**HWp**), **Optional (it gives bonus points)**

The final grade formula is:

$$\text{ROUND}[\text{written_test} + (\text{HW or HWp})]$$

An outcome is passed if grade $\geq$ **18**. If not passed, then we cannot record your grade.

So to record your grade: written test $\geq$ **18**

1. Written test:

The written test is in the form of:

- open or closed questions
- show computation on a small toy problem to demonstrate you are comfortable in applying concepts (aka exercise)
- In some cases: mathematical definition, proof sketch.
- No programming skills are required, but at least for part II, sporadically, we may post code in Python explained in class, and ask to describe what it does in the context of NLP.

Questions will follow the [format to this exam of machine learning](#), but obviously concerning Natural Language Processing. **The exam is in person only**, per Sapienza regulations. The written test grants you a maximum of **28** points. The exam is a written test of about **90 minutes** (1.5 hours).

2. Homework (HW) or HW extended as a project (HWp)

The homework is optional and meant to let you have some “fun” with NLP tools and datasets, and demonstrate that you are comfortable with using terminology of NLP, tools used in NLP, NLP tasks, and prominent datasets. The homework is a report on the state of the art in NLP. We will release a LaTeX template with instructions. In the HW, you would freely select **an NLP task** of interest among those presented during the course.

 Changes from previous year:

- **New:** This academic year, the HW (or HWp) is **optional**. We have a new grading system based on a max of 28 points for the mandatory written test + bonus points, optional with HWp.
- The LaTeX template will be used only to submit your HW/HWp report. To submit the proposal, we will use a quicker Google Form.
- **Added statement about the use of GenAI tools**

Given the task, you have to:

- Describe the NLP task
- Provide examples of input and expected output
- What is the current state-of-the-art for this task (study related work or surveys)
- Applications for this task
- List the datasets available for
- Existing tools or libraries that can be used
- Papers with code
- State-of-the-art evaluation

The timeline for the HW/HWp:

1. You make a proposal (by filling out the following Google form:

<https://forms.gle/9XSy3DvZbMiVvT7C9>

2. about the NLP task that you are going to address in the HW/HWp
3. We will agree on your proposal or send you the proposal back with suggestions. We repeat reviewing until agreement (**note our reviewing process and your proposal is part of the evaluation!**).
 - a.  For "agreeing on you proposal", what we will do is that we will keep an updated published and anonymized google sheet with all the proposal and the one that are validated will appear with a green symbol like . From that moment, you can start working on the proposal. We will not update you on our side when to start, so please check the google sheet every other day for possible updates.
 - b. Instead in case the proposal needs changed we will write to you on classroom with the request and add a sign like  in the google sheet.
4. After you see  in the doc , you will start to work on the HW.

5. You send the HW as a final PDF file (or ZIP file if necessary), before each exam deadline.

The HW evaluation is as follows: at best, if you submit the HW only, you may get **2 additional** points.

The HW can be extended into a project (**HWp**) if you add YOUR OWN comparative evaluation of at least Y existing methodologies against at least a baseline on a selection of Y datasets with your experimental results and a scientific discussion. You will release the ZIP file with code. Where Y is the number of team members.

Teaming up: The HWp are meant to be done as a single student yet teaming up may be allowed in some cases, **up to a maximum of 2**. Anyway, if you team up, at the end of your report you have to insert a contribution statement declaring which author did what.

Use of GenAI and LLM tools: Students are required to write a statement of the usage they used in the report and in their experimentation with Generative AI tools such as LLMs, as well as Vibe Coding, Codex, etc. Please be as specific as possible on which tools you use and why. We hope to discourage situations where we find a citation that is completely invented by an LLM. **You can use these tools, but it is your primary responsibility for the content and material that you submit.** If we find that you used some of these tools in ways you did not disclose, it may affect your grade.

A team of Y members will deliver HWp of at least $Y*2$ pages (references excluded).

The HWp evaluation is as follows: at best, if you submit the HWp you may get **4** points.